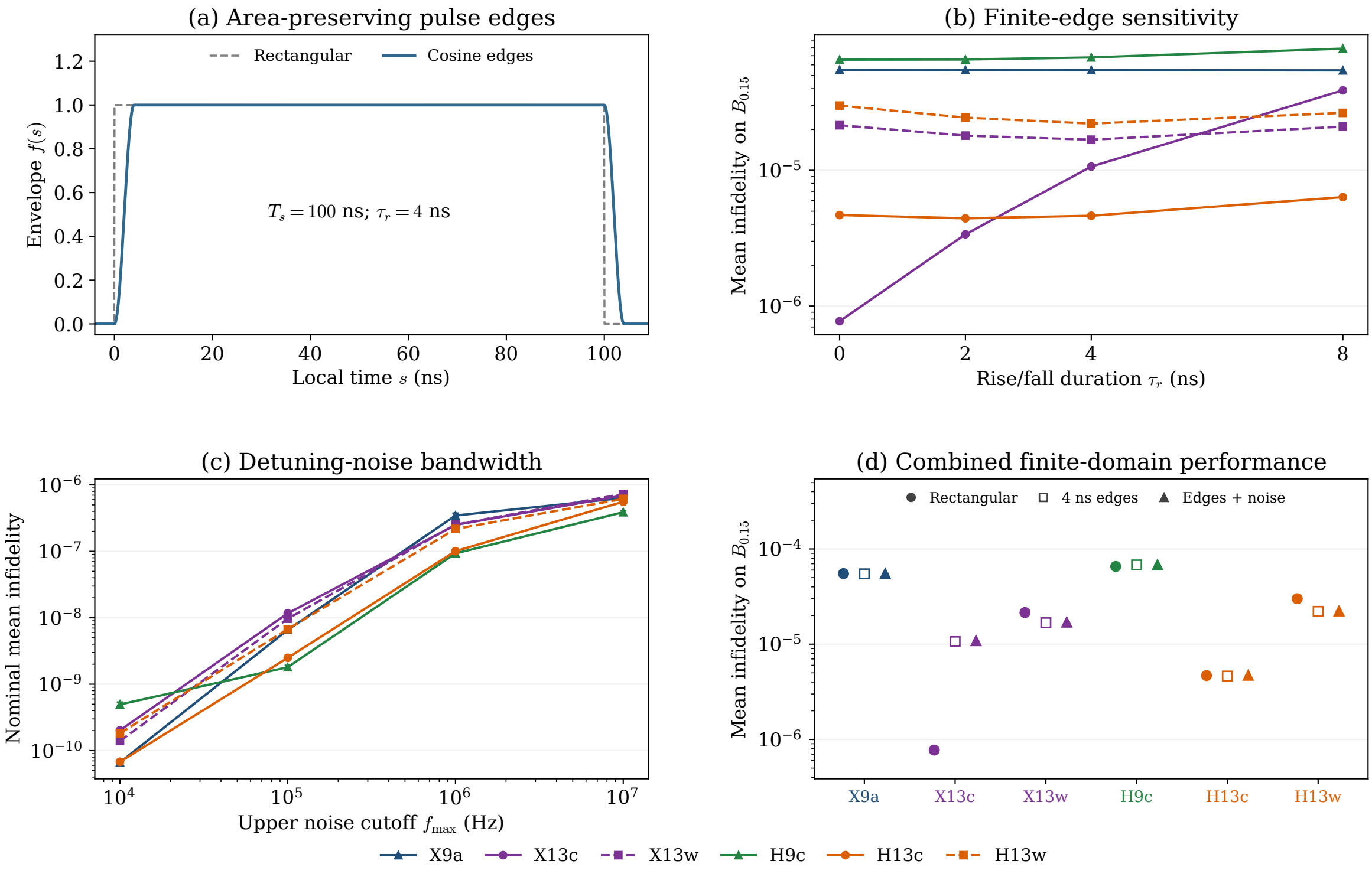
(a) Area-preserving pulse edges
Rectangular
Cosine edges
Envelope $f(s)$
$T_s = 100$ ns; $\tau_r = 4$ ns
Local time s (ns)
(b) Finite-edge sensitivity
Mean infidelity on $B_{0.15}$
Rise/fall duration τ_r (ns)
(c) Detuning-noise bandwidth
Nominal mean infidelity
Upper noise cutoff $f_{\max}$ (Hz)
(d) Combined finite-domain performance
Rectangular
4 ns edges
Edges + noise
Mean infidelity on $B_{0.15}$
X9a
X13c
X13w
H9c
H13c
H13w